Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Smart-Clearance

Every near-expiry carton gets a second chance, chosen by AI.

a. Team name: **Silicon Prophets**

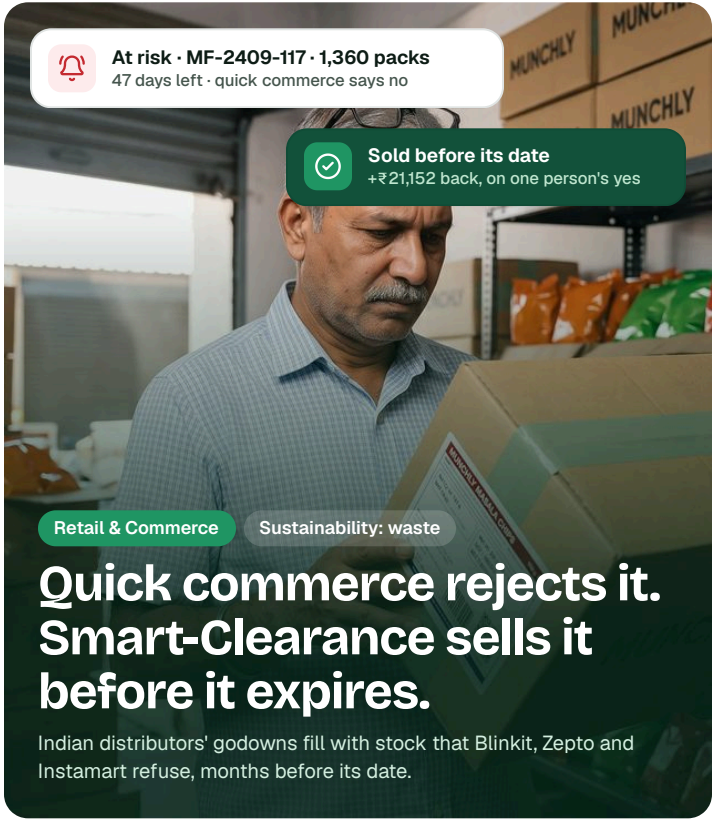
b. Team leader name: **Shruti Harindran** with Arun Kumar Dutta and Vignesh Nagarajan

c. Problem Statement: **Retail & Commerce: Intelligent Customer and Business Experiences**

Inventory intelligence and demand planning for FMCG brands: AI agents find near-expiry stock in distributors' godowns and route each batch to the exits that recover the most before its date, with one person's yes.

Brief about the idea

Smart-Clearance Silicon Prophets



THE PROBLEM

FSSAI (Nov 2024): food sold online must reach the buyer with 30% of its life, or 45 days, left. Blinkit, Zepto and Instamart inward only with more, 60 to 90 days or about 60% of its life, so FMCG brands' stock piles up in distributors' godowns **months before its date**. Most is destroyed: written off, GST credit reversed, disposal and EPR paid, the claim settled.

THE IDEA

Agentic AI, sold as SaaS to FMCG brands. For each near-expiry batch, ten agents on Gemini work out which of India's own channels recover the most before the date (nearby kiranas, marketplace buyers, a staff sale, a food bank), then do the work: the offers, the negotiation, the GST papers. **One person approves the plan**, with the money on screen.

HOW IT WORKS · ONE BATCH, FROM THE SHELF TO THE LEDGER

Masala Chips MF-2409-117, Rakesh Traders, Nagpur



1 DETECT · Watcher · no model

Flagged while there is still time

Sell-through from BigQuery, projected to the day each app stops taking it: 1,360 of 1,840 packs won't sell in time.



2 VERIFY · Vision · Gemini Flash

The carton is the truth

Rakesh photographs the label; Gemini reads batch, dates and MRP, and backend-api checks them against the stock record.



3 DECIDE · Valuer, Router · Gemini Pro

Every exit priced, one yes

Each exit priced net of costs and GST against the bin; the Router splits the batch; Priya sees ₹21,770 planned and says yes.



4 EXECUTE · Outreach, Lister, Negotiator

The agents do the selling

38 nearby kiranas get a Hindi scheme and 31 order 588; 772 packs go on ExpireSoon, where a ₹13 bid is countered to ₹14.20.



5 SETTLE · Paperwork · Gemini Flash

The papers, signed and stamped

Tally tax invoice with IRN and signed QR, a GST invoice per kirana, and an ₹8,768 credit note that leaves Rakesh whole.



6 REPORT · Impact · Gemini Flash

One ledger, from the papers

Impact posts money recovered, GST credit kept, kilos out of landfill, CO_2e and the BRSR line, each backed by a paper.

FIVE EXITS, MIXED BATCH BY BATCH · THE ROUTER PICKS THE SPLIT THAT RECOVERS THE MOST

Kirana scheme · Marketplace lot · Staff sale · Food bank · Destroyed, on proof

Shops nearby get an offer in their own language, buyers outside the brand's territories bid on a lot, staff buy at the godown by UPI, a food bank collects a legal donation; only what no channel took is destroyed, on proof. Every person shown is fictional.

Opportunities

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Everyone else does one piece. Smart-Clearance decides, then does all of it.

1 How different is it from any of the other existing ideas?

Can it…	Stock and sales apps Bizom, FieldAssist	Near-expiry marketplaces NearXpiry, EOL Stocks	Kirana and resale networks Udaan, Kirana Club, GoFig	Food banks Feeding India, IFBN	Smart-Clearance
Spots the batch at risk early, per app's gate	◐	○	○	○	●
Verifies it from the shelf: the label, read by AI	○	◐	○	○	●
Prices every exit against the bin, GST included	○	○	○	○	●
Splits one batch across many exits	○	○	○	○	●
Does the selling: lists, Hindi offers, negotiates	○	◐	◐	○	●
Drafts GST papers, FSSAI checklist, BRSR line	◐	○	○	◐	●
Leaves the distributor whole, in his name	○	○	○	○	●

● yes ◐ in part ○ no · our market scan, Oct 2026. Abroad, Spoiler Alert serves big US liquidators; India has only lakhs of kiranas.

2 How will it be able to solve the problem?

Watch → Verify → Price → Split → One yes → Execute → Settle

Found in time, proven from the carton, priced against the bin, then sold and papered by agents after one yes. **Why it works now, in India:**

01
The law and the apps set the expiry clock
FSSAI: 30% of life or 45 days left at delivery. Apps inward at 60 to 90 days or ~60%.

02
In India, the bin is taxed
Destroying reverses the GST credit (CGST s.17(5)(h)) and adds disposal and EPR.

03
Waste is now on the record
SEBI's BRSR asks for waste; FSSAI's 2019 rules let food be donated before expiry.

04
Bharat's trade is reachable
Shops run on smartphones and UPI; Gemini reads a label and writes a Hindi offer.

Who gains: the brand (money back, its GST credit, a BRSR line); the distributor (whole, no claim tussle); the kirana (₹266 on 24 packets, his own GST invoice); food banks and the planet. All but the brand use it free.

3 USP of the proposed solution

Routes, not lists
+₹21,152 instead of −₹26,330 on one batch
Prices every exit net of its costs and GST against the bin, then splits the batch under each channel's demand, a territory guard and the days left.

It reads the real world
72 of 73 label photos read right
Gemini reads the carton's printed label, a UPI receipt and a destruction's photos; backend-api checks every read against the record.

Money a CFO trusts
₹1,224 of input GST kept on one batch
Every rupee is computed in code, one set of rules in JS, TS and Python held by fixtures; the models only write the words.

Papers that stand up to audit
Append-only audit log, a trace per request
Tax invoice with e-invoice IRN and signed QR, credit notes, the ITC memo, a GST invoice per kirana order, signed certificates.

Built for Bharat's trade
Nothing to install: it opens in the browser
The distributor acts from his phone and his own Tally series; kiranas read the scheme in their language; money moves by UPI.

Agents do it, a person decides
10 agents · 2 human gates
Nothing is listed, messaged or shipped before one yes, and destroying stock needs a second. Each brand sets suggest, ask or act.

Impact with evidence
217.6 kg and 544 kg CO_2e from one batch
Kilos resold, donated and destroyed, CO_2e, meals and EPR plastic, counted from the papers; the BRSR Principle 6 table exports as is.

One platform, any manufacturer
₹1.70 of Gemini for a batch's journey
A workspace per brand from its supply-chain profile; distributors, kiranas, buyers and food banks use it free.

List of features offered by the solution

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DETECT · DATA, WATCHER
Batches at risk, early
Each night the Data agent loads every distributor's stock export into BigQuery; each morning the Watcher projects sell-through to the day each quick-commerce app stops taking the batch. The chips: 1,360 packs flagged, 47 days left.

Model: Gemini Flash | rules, no model

VERIFY · VISION
The label is the truth
The distributor takes or uploads one photo of the carton. Gemini reads batch, dates, MRP and pack as printed, never from the barcode; the read is checked against the stock record before anything is priced.

Model: Gemini Flash

VALUE · VALUER
Every exit priced
Each exit's recovery is computed in code against the bin's true cost: the stock, the GST credit reversed, disposal and EPR. Gemini Pro notes why each exit suits this batch. The chips' bin: −₹26,330.

Model: Gemini Pro

ROUTE · ROUTER
One batch, split
The mix that recovers the most under each channel's demand, a territory guard and the days left, explained in plain words. The chips: 588 packs to kiranas, 772 to a buyer outside Munchly's territories.

Model: Gemini Pro



APPROVE · A PERSON
One tap releases it
The plan, what it recovers and the bin it avoids are on screen before the yes. Nothing is listed, messaged or shipped until then; the yes is kept in the approver's name on the batch's Record.

Model: a person's yes

EXECUTE · OUTREACH
Kiranas, in their language
A scheme in Hindi, English or Marathi reaches the shops that already sell the SKU; each orders in one tap and gets a GST invoice; the van round follows the orders. 31 of 38 shops ordered 588.

Model: Gemini Flash

EXECUTE · LISTER, NEGOTIATOR
A marketplace, negotiated
The lot is listed on ExpireSoon in the distributor's name, hidden inside the brand's territories. Bids are countered to a reserve no model ever sees: a ₹13 bid closed at ₹14.20.

Model: Gemini Flash | Gemini Pro

EXECUTE · DONATION, STAFF SALE
Every pack finds a home
A food bank collects what shops can't sell in time, with its FSSAI checklist, a receipt and the meals counted. A staff sale at the godown is paid by UPI and proven by one receipt Vision reads.

Model: rules | Gemini Flash

SETTLE · PAPERWORK
GST papers, signed
The tax invoice with its e-invoice IRN and signed QR, the e-way bill check, credit notes, the ITC memo, an invoice per kirana order and any destruction certificate: signed, stamped PDFs.

Model: Gemini Flash

SETTLE · EXPIRY
Leftovers, settled
On the best-before, packs no channel took settle by the client's own policy. Destroyed at the godown, they need two photos, Vision's checks, an agency's certificate signed in place and a second yes.

Model: Gemini Flash | a second yes

REPORT · IMPACT
One ledger
Money, GST and impact per batch, quarter and year: kilos resold, donated and destroyed, CO_2e, meals and EPR plastic. BRSR Principle 6 and GST summaries export as CSV; every batch keeps its Record.

Model: Gemini Flash

PLATFORM · PORTALS, CONSOLE
Every party, its own view
Distributors, kiranas and food banks each see their own batches, orders and papers. Staff onboard a client in seven steps and set how far each agent goes: Suggest, Ask or Act.

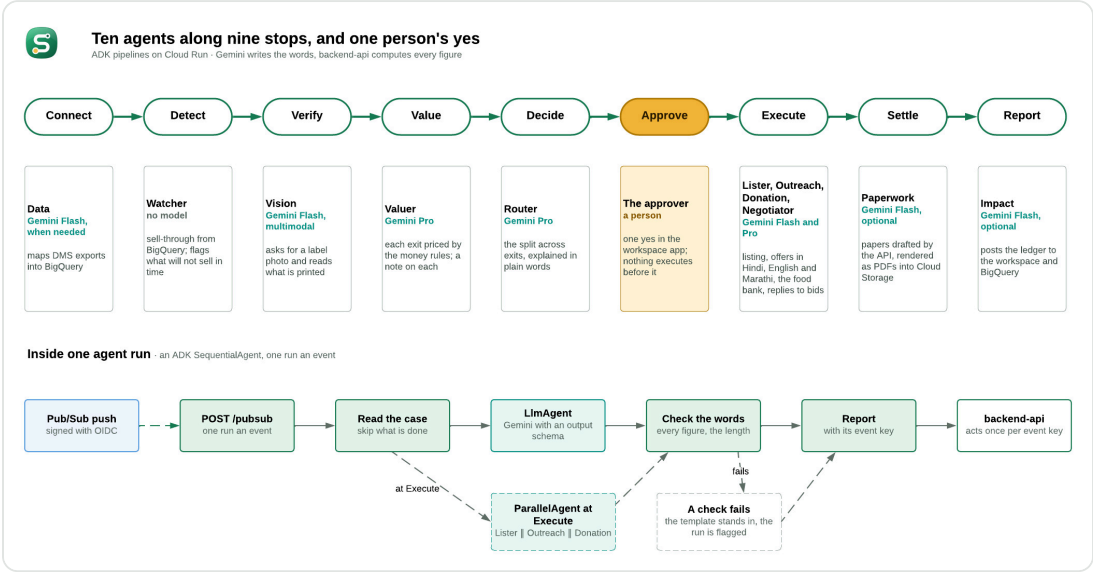
Model: no model

Teal: an agent on Gemini (Pro decides, Flash reads and drafts, rules where the sum must be exact). **Amber**: a person's yes. **Violet**: the marketplace. Live everywhere: server-sent events move every screen, Firebase Cloud Messaging pushes to phones, and the workspace installs as a PWA.

Process flow diagram or Use-case diagram

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PROCESS FLOW · ONE BATCH THROUGH NINE STOPS



LUCIDCHART Smart-Clearance Architecture on Google Cloud, page 2: ten agents along nine stops, and one person's yes

1. **Nine stops, left to right:** Connect the stock data, Detect, Verify the label, Value each exit, Decide the split, Approve, Execute, Settle the papers, Report.
2. **Who works at each stop:** the card under a stop names its agent and model: Pro where a person acts on the words, Flash for reading and drafting, no model where the sum must be exact.
3. **One human stop:** Approve is amber. Until a person sees the plan and the money and says yes, nothing is listed, messaged or shipped.
4. **Inside one agent run:** a Pub/Sub event starts one run on Cloud Run; it reads the case, asks Gemini for words in a fixed schema, checks every figure, and reports once, keyed so a retry never acts twice. If a check fails, a template stands in.

USE CASES · WHO DOES WHAT



Neha · Smart-Clearance staff · console

Onboard a client in seven steps | Tune each agent: suggest, ask, act | Reset the journey, fire runs



Priya · brand's supply chain · approver

Confirm Setup | Approve the plan | Review the signed papers | Second yes on a destruction | Read the ledger



Rakesh · distributor · Nagpur

Allow the agents once | Send the label photo | Issue the invoice from Tally | Run the van round | Destroy leftovers, on proof



Lakshmi Agencies · distributor · Hyderabad

Record a staff sale on its UPI receipt | Read her papers and credit



Ganesh ji · kirana · one of 38 shops

Order from the Hindi scheme | Get his own GST invoice | Decline: Not this time



Agrawal ji · marketplace buyer · Raipur

Bid on the lot | Accept the counter, pay the token



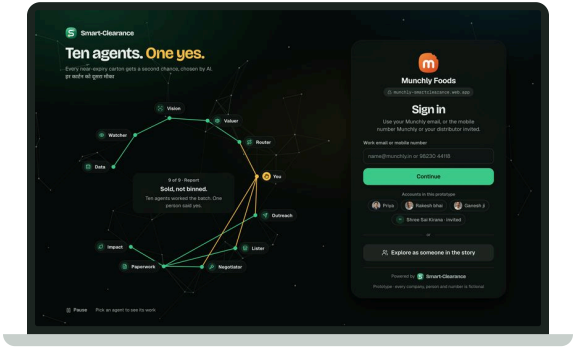
Meera · food bank · Feeding India

Collect the donation | Issue the receipt, count the meals

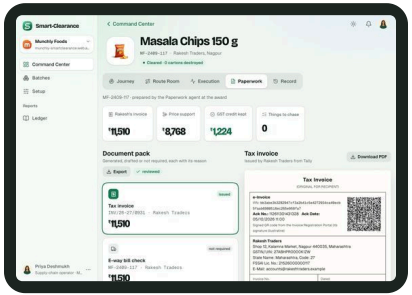
amber a person's yes; the agents do everything else. Every person and company here is fictional.

Wireframes/Mock diagrams of the proposed solution (optional)

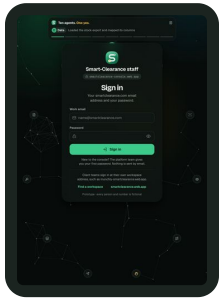
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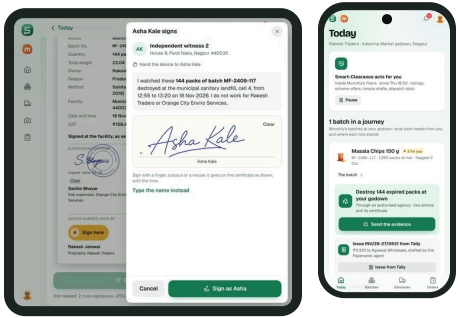
Laptop · the workspace sign-in DARK
"Ten agents. One yes.": the agents work a batch behind the form



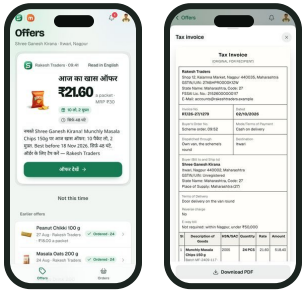
iPad, landscape · Priya's papers
the tax invoice with its e-invoice IRN and signed QR



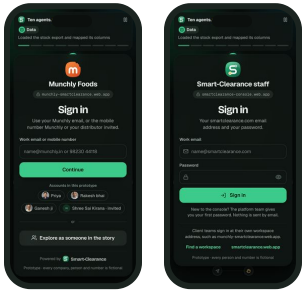
iPad · the console DARK
staff sign in



Pixel Fold · open, then folded
the signing pad, each witness signs in place; Rakesh's Today: destroy 144 packs



iPhone · a kirana
the Hindi offer, ₹21.60 a packet; his tax invoice, ₹454



iPhone · signing in DARK
Munchly's workspace; the staff console

Progressive web app — **One installable app, one codebase, every screen, light and dark**

Installs from the browser
Add to Home Screen on Android, iOS and the desktop: its own icon, full screen, no app store.

Light or dark
Every screen in both themes, following the device or the person's choice.

The device's own camera
Label, evidence and receipt photos in the browser; the signing pad takes a finger, stylus or mouse.

One codebase, every screen
SvelteKit lays the same screens out from 390 to 1,440 px: laptop, tablet, foldable, phone.

Next · native apps
iOS and Android on the same API: offline capture at the godown, background sync.

Design first
Every screen began as two or three options on a review board (45+ rounds); only the pick was built.

Architecture diagram of the proposed solution

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People and apps
Firebase · web and phone

- **Priya** — operator · approves
- **Rakesh bhai** — distributor
- **Ganesh ji** — kirana
- **Agrawal ji** — marketplace buyer
- **Meera** — food bank
- **Neha** — staff · console

Firebase Hosting (four apps), Auth and Cloud Messaging

One person's yes releases the agents

backend-api
FastAPI on Cloud Run

- the only writer of the record
- computes every rupee, split and paper
- cuts each view to the person's role
- a change, its audit line and its events commit together

Pub/Sub
events after commit, by batch

facts in, words out (/internal, Google ID token)

Cloud SQL
PostgreSQL 18: the record, an append-only audit log

Cloud Scheduler
calls backend-api's tick every minute

daily runs, due timers, stalled events

Agents service
Google Agent Development Kit on Cloud Run · one pipeline per event

Data	Flash	Watcher	rules
Vision	Flash	Valuer	Pro
Router	Pro	Lister	Flash
Outreach	Flash	Negotiator	Pro
Paperwork	Flash	Impact	Flash

INSIDE EVERY RUN

read the case › LlmAgent + schema › check the words › report once

The reserve never enters a prompt; a template stands in for any reply that fails its check.

Gemini
on Agent Platform, global endpoint

Gemini Pro

judgement: the Valuer, Router, Negotiator's bids

Gemini Flash

reading and drafting, labels from photos

structured output; no API key

BigQuery
sell-through, prices, the ledger, every run

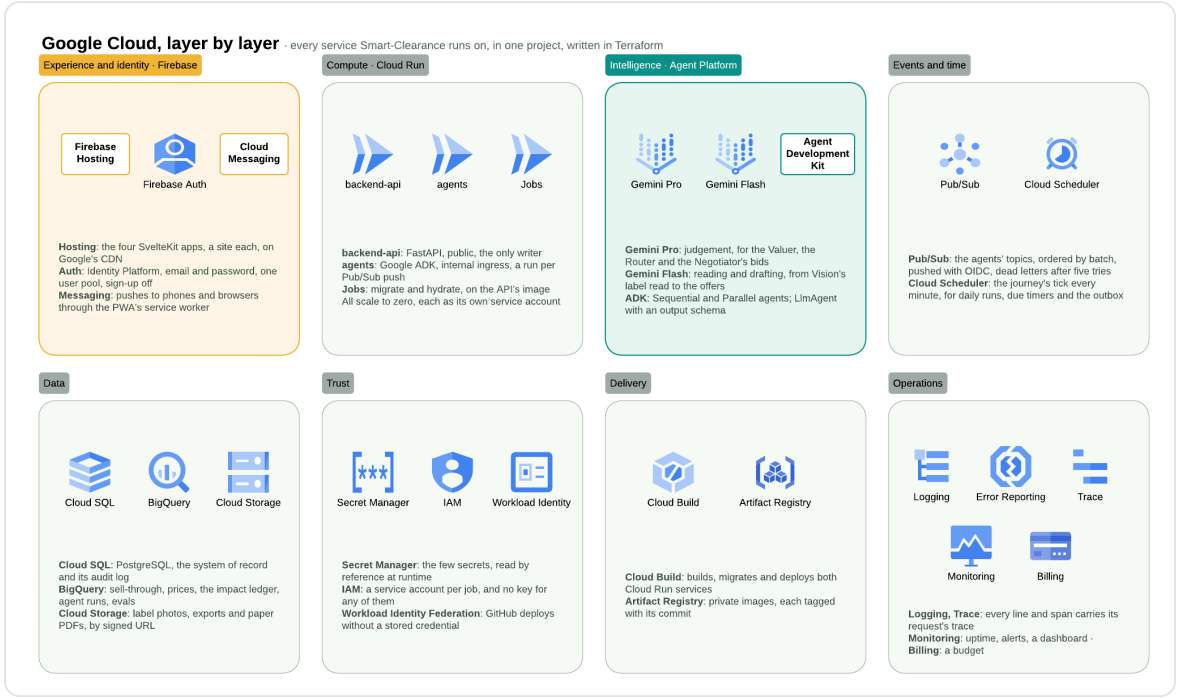
Cloud Storage
label photos, exports, PDFs by signed link

TRUST AND OPERATIONS · KEYLESS, TRACED, IN TERRAFORM

Secret Manager · IAM, a service account a job · Workload Identity · Cloud Build · Artifact Registry · Cloud Logging · Cloud Trace · Cloud Monitoring · Billing budget

Technologies to be used in the solution

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LUCIDCHART Smart-Clearance Google Cloud and AI, page 1: Google Cloud, layer by layer; every service in one project, written in Terraform

GOOGLE FIRST

Google AI

Gemini 3.1 Pro (preview) and Gemini 3.8 Flash on Agent Platform · Google Agent Development Kit 2.11 · the google-genai SDK

Google Cloud and Firebase

Cloud Run · Cloud SQL · BigQuery · Pub/Sub · Cloud Scheduler · Cloud Storage · Cloud Build · Secret Manager · Cloud Trace · Firebase Hosting, Auth and Messaging

Frontend

SvelteKit 3 · Svelte 5 · TypeScript 6 · Vite 8 · Tailwind 4 · an installable PWA

Backend

Python 3.14 · FastAPI · SQLAlchemy · PostgreSQL 18 · OpenTelemetry

Delivery and quality

Terraform on Google Cloud · GitHub Actions with Workload Identity Federation · Playwright · axe-core · Vitest · pytest

Generative media, for the demo films

Veo 3.1 · Nano Banana 2 (Gemini 3.1 Flash Image) · Gemini-TTS · Lyria 2, on Agent Platform

The track's suggested stack, used in full: Gemini on Agent Platform, BigQuery, Cloud Run, Cloud Storage and Cloud SQL, plus Firebase, Pub/Sub and Cloud Scheduler.

Estimated implementation cost (optional)

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RUNNING IT ON GOOGLE CLOUD, A MONTH, IN RUPEES

SERVICE	WHAT RUNS, AT GOOGLE'S LIST PRICE	₹ A MONTH
Cloud SQL for PostgreSQL 18	db-f1-micro (shared vCPU, 0.6 GB) at ₹1.21 an hour, zonal; 10 GB SSD, backups, point-in-time recovery	₹1,115
Cloud Run	backend-api (1 vCPU, 512 MiB) and the agents (1 vCPU, 1 GiB), 0 to 2 instances, billed per request; most of it is the live stream while people use the app	₹100 – 600
Gemini on Agent Platform	Flash at ₹144 / ₹720 a million tokens in / out, Pro at ₹192 / ₹1,152; about ₹2 a batch's journey (light / heavy use)	₹130 – 1,900
Pub/Sub, Cloud Scheduler, BigQuery	events, the minute tick, the analytics tables	₹0
Firebase Hosting, Auth, Cloud Messaging	four sites, one user pool, pushes	₹0
Secret Manager, Cloud Build, Trace, Logging, Monitoring	secrets, releases, spans, JSON logs, alerts	₹0
In all	**at the prototype's traffic, light use to heavy use**	**₹1,350 – 3,600**

Google's Cloud Billing catalogue in INR for asia-south1 (Mumbai), 10 Oct 2026. A budget of about ₹3,840 emails at 50%, 90% and 100%. A full eval run on live Gemini, about ₹260, runs on request only.

FREE AND OPEN SOURCE, END TO END

Frontend	SvelteKit · Svelte 5 · Vite · TypeScript · Tailwind
Backend	FastAPI · SQLAlchemy · Alembic · PostgreSQL 18
Agents	Google ADK (Apache 2.0) · google-genai · Jinja2 · WeasyPrint
Quality	Playwright · axe-core · Vitest · pytest · OpenTelemetry
Look	Lucide icons · Geist, Bricolage and Noto fonts (OFL)

No licence fees anywhere in the stack; Terraform's CLI is free to use.

GOOGLE'S FREE TIERS WE RUN INSIDE

- Cloud Run: 2 million requests and 180,000 vCPU-seconds a month
- BigQuery: 1 TiB of queries and 10 GiB stored a month; Pub/Sub: 10 GiB of messages
- Cloud Scheduler: 3 jobs (we run 2); Secret Manager: 6 secret versions
- Cloud Trace: 2.5 million spans; Logging: 50 GiB; Cloud Build: 2,500 build-minutes
- Firebase Hosting, email sign-in and Cloud Messaging; GitHub Actions on a public repository

ONE BATCH, END TO END

₹1.70
of Gemini for one chips journey, on production

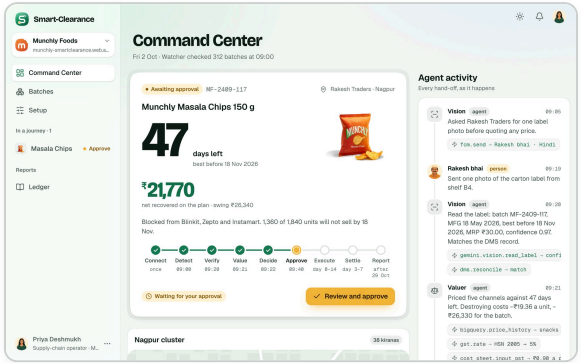
₹21,152
recovered on that batch, not written off

0.01%
the AI's share of what it recovers

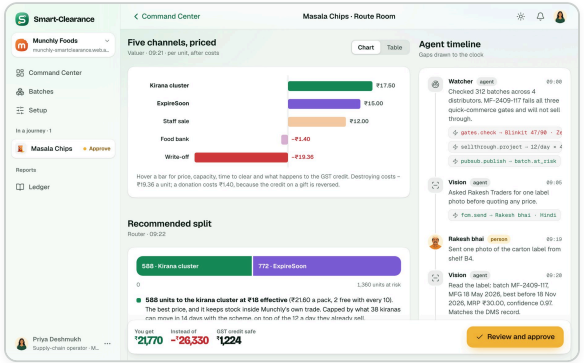
Snapshots of the prototype

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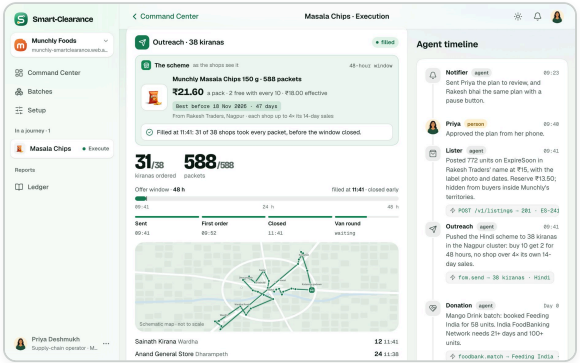
THE OPERATOR'S DAY · PRIYA, MUNCHLY'S SUPPLY CHAIN, ON ONE BATCH OF MASALA CHIPS



Command Center · Operator
Priya's home: the batch at risk, its days left, the plan's ₹21,770, the nine stops, each agent's hand-off, and Review and approve.



Route Room · Operator
Five channels priced per unit after costs, against the write-off, then the Router's split: 588 to kiranas, 772 to ExpireSoon.

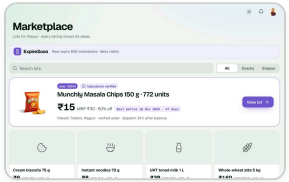


Execution · Operator
The kiranas' Hindi scheme filled: 31 of 38 shops, 588 packets in 48 hours, the van round drawn from their orders, the agents' timeline.

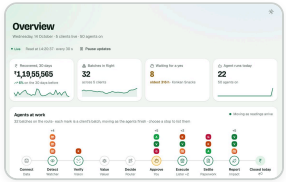
EVERY OTHER PARTY, THE PAPERS AND THE PROOF



Ledger · Finance and ESG
Money, GST and impact by batch, quarter and year: ₹2.42 lakh recovered this year.



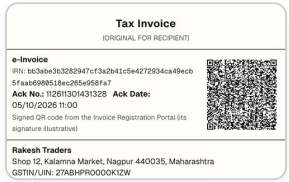
ExpireSoon · Buyer
Agrawal ji sees the lot outside Munchly's territories, 50% off MRP, and bids; the Negotiator counters.



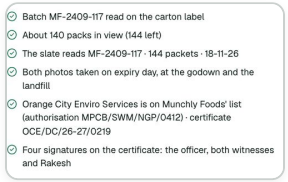
Console overview · Staff
Every client at a glance: recovered, in flight, waiting for a yes, the agents at work stop by stop.



Label photo · Distributor
The carton's statutory label; Gemini reads batch, dates and MRP, matched to the stock record.



Tax invoice · Paper
Issued from Tally with its e-invoice IRN and signed QR, every party's GSTIN, signed and stamped.



Vision's four checks · Operator
Before a destruction's second yes, Gemini checks the batch, the count, the slate and the agency.

The workspace app and the console at commit 0855b78 (11 Oct 2026), on the sample journey. Munchly Foods and every person, number and paper in them are fictional.

Prototype Performance report/Benchmarking

Smart-Clearance Silicon Prophets

77 / 77
steps passed in the recorded run on production, reset to impact report, in 9 min 46 s

0
WCAG 2.2 AA violations across the four apps' builds, in 352 checks

1,199
automated tests pass: 547 backend, 202 agents, 450 frontend

2.6 s
first paint on a 3G line for the console, from 8.1 s

102 ms
median API call on Cloud Run; 412 ms at p95, over 7,251 calls

8.7 s
median agent decision on Gemini Pro; Flash steps take 2 to 4 s

₹0.06–0.50
of Gemini a call; ₹1.70 for a whole chips journey

27%
peak CPU on Cloud SQL's smallest tier, db-f1-micro; 11% average

THE AGENTS ON LIVE GEMINI · EVAL RUN 2, 208 CASES

Vision · label read	72/73 · 98.6%
Data · column map	25/25 · 100%
Router · the split	20/20 · 100%
Lister · listing	15/15 · 100%
Outreach · offers	15/15 · 100%
Negotiator · bids, chat	38/40 · 95%
Valuer · exit notes	19/20 · 95%

A Gemini Pro judge scored the Router 4.61 and the Valuer 5 out of 5; the reserve price never leaked in 40 negotiation cases. The three misses were the Pro preview's quota (429), each falling back safely to a template.

How an eval runs: fixed cases, each with its one right answer in the file · run through the agent's production pipeline on live Gemini · scored in code (exact fields, no leak, only the plan's figures) · words judged by Gemini Pro against a rubric · each run's summary committed and replayed in CI with no live model.

EACH AGENT ON PRODUCTION · P50 · P90 · ₹ A CALL

Paperwork Flash	2.1 · 5 s	₹0.06
Lister Flash	2.5 · 17 s	₹0.15
Outreach Flash	3 · 9.2 s	₹0.22
Vision Flash	4 · 48.6 s	₹0.34
Negotiator Pro	5.6 · 14.9 s	₹0.15
Valuer Pro	9.2 · 26.6 s	₹0.50
Router Pro	10.8 · 38.4 s	₹0.26

median · 90th percentile, with retries

784 agent runs and 257 Gemini calls, 8 to 10 Oct 2026, read from BigQuery. Flash does the volume; Pro only the decisions a person approves. Each call gets up to 3 attempts with back-off, 30 s each.

How it scales: each step is one Terraform variable. Cloud Run to 20 API instances and 10 agent instances, ₹0 until used; Cloud SQL from db-f1-micro (₹1,115) to db-g1-small (₹3,200) or 2 vCPU and 8 GB (₹11,900); Provisioned Throughput for Gemini, so no 429s.

BENCHMARKS

MEASURE	RESULT
Parity with the approved designs	305 pixel comparisons pass, within 0.5%
Landing page on 3G	first paint 6.4 s → 2.4 s; loader lifted 14.6 s → 7.8 s
Journey suites	chips, mango and both leftover scenarios pass, each recorded
Money rules	one set in JS, TS and Python, held to the same answers by fixtures
Evals in CI	every pull request replays the agents' recorded answers, with no live model

Measured on production (Cloud Run, Cloud SQL, Agent Platform, asia-south1) and on the builds; Google's INR list prices, 10 Oct 2026.

The demo films, made on Agent Platform: Veo 3.1 scenes at 1080p, image to video, SynthID in every frame; Nano Banana 2 cast sheets and 2K keyframes; Gemini-TTS narration and a Lyria 2 score; each call priced first, about US$92 in all.

ONE TRACE IN PRODUCTION, 10 OCT 2026

`134976f7785da3621103263c699520d3` 7.28 s · 368 spans, from Priya's tap to the agents' work

Priya taps Approve	0.77 s	Outreach · Gemini 3.8 Flash	2.40 s
agents · execute	6.89 s	Outreach reports the scheme	3.47 s
load settings and the case	0.81 s	Lister reports the listing	1.13 s
Lister · Gemini 3.8 Flash	2.76 s	BigQuery · agent_runs	0.17 s

One id, everywhere it went: Cloud Trace (331 spans from backend-api, 32 from the agents, each Gemini call with its model and tokens), Cloud Logging (7 lines from both services), BigQuery (each agent run's row) and Cloud SQL (Priya's audit line, “approved the plan”). The apps send a W3C traceparent; OpenTelemetry and Google ADK carry it through Pub/Sub.

Additional Details/Future Development (if any)

Smart-Clearance Silicon Prophets

BUILT SO FAR

184
Jira issues, from the first prototype to production

4
apps live on Firebase Hosting, one installable PWA

10
agents live in production on Cloud Run

45
design rounds, each picked from options

Built to be trusted: WCAG 2.2 AA on every app, one trace per request, every Google Cloud resource in Terraform, and no service-account keys.

EVERY STEP ON THE RECORD

- **In their own name:** each change commits with its audit line: who, for whom, what, which batch, when, the trace id.
- **Agents in their own words:** each run sits on the same trail, in plain words.
- **Only people say yes:** the plan, the papers and a destruction, each kept in its giver's name.
- **Cannot be rewritten:** Postgres grants and triggers refuse any update or delete; the trail downloads as CSV.

ONE PLATFORM, MANY MANUFACTURERS

A new client in seven steps, from its supply-chain profile, in the console:

1 Company · 2 Workspace · 3 Supply chain · 4 Exits · 5 Agents · 6 People · 7 Review

Staff then upload its first stock export and the Data agent maps it; every change is audited.

How far each agent goes, a preset per client, then agent by agent; the approval is never switched off:

Cautious	Standard	Trusted
Nilgiri Naturals · cosmetics	Munchly Foods · snacks, drinks	Kaveri Dairy · dairy
Data and the Watcher act; the rest suggest. No food bank for cosmetics.	Most agents act; Vision, the Negotiator and Paperwork ask first.	Agents act; the Negotiator and Paperwork ask. No marketplace for short-life dairy.

Each client's own: gates per SKU with per-batch overrides, the Negotiator's hidden reserve (₹13.50 a pack for Munchly), the leftovers policy and its agencies, 90 days of sell-through, and the kiranas each distributor serves (38 around Nagpur, 58 around Hyderabad).

WHAT COMES NEXT

INTEGRATE

- **WhatsApp Business** for kirana offers beside the in-app push
- **Partner APIs** for near-expiry marketplaces such as NearXpiry
- **DMS connectors** (Bizom, Tally) instead of nightly CSV exports
- **Live e-invoicing** (IRN) and e-way bills through a GST suvidha provider

DEEPEN THE AI

- **Demand forecasting** per pincode with BigQuery ML and Agent Platform
- **Gemini Pro on GA**, and Tamil, Telugu, Bengali and Kannada
- **Full-credit take-back** beside destruction at the godown
- **Agents that learn** prices and splits from every closed deal

SCALE THE SAAS

- **Native iOS and Android apps**: offline capture, background sync
- **Self-serve onboarding** for manufacturers
- **Own domain and single sign-on**; data residency for Enterprise
- **Pharma and OTC profiles** under the Drugs and Cosmetics Rules

Proven before a client's first batch: each agent passes its fixed eval set (208 cases in seven sets, on live Gemini) before it touches a client's batch, and every pull request replays the recorded answers in CI.

NEXT UP · REJECT AND RE-PLAN

today the operator approves a plan or leaves it (Reject is on screen, marked Coming Soon); next, a rejection with a reason brings new options, each kept in the audit log

The plan
the Valuer prices each exit, the Router splits the batch

→

Reject, with a reason
“no ExpireSoon for this brand”, “keep the price above ₹14”

→

Re-plan
the agents take the reason as a rule, re-price and re-split

→

New options
two or three plans, the money shown for each, back for one yes

Provide links to your:

Smart-Clearance Silicon Prophets

GitHub Public Repository
github.com/aibuilder2026/smart-clearance
The frontend, backend-api, the agents, Terraform, the designs and every decision record.

Demo Video Link (3 Minutes)
youtu.be/5eyNqJ3zhJQ
The working demo, recorded live on production: the console, the Mango Drink sold through every exit, and the Masala Chips destroyed against evidence, with every person's screen.

Final Product Link

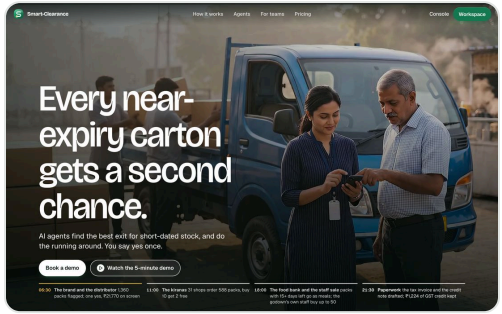
THE CLIENT WORKSPACE
munchly-smartclearance.web.app
Munchly Foods' workspace, live on Cloud Run and Gemini (fictional demo client)

THE GUIDED DEMO
smartclearance-demo.web.app
the whole journey, beat by beat, with no sign-in

THE LANDING PAGE
smartclearance.web.app
the platform's own site; its bar leads to the Console and the Workspace

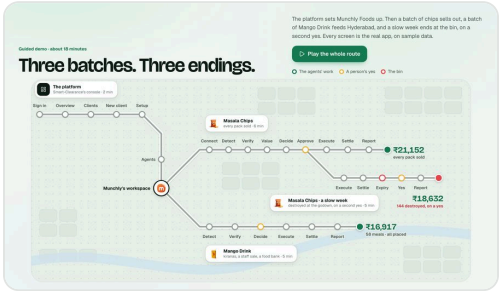
THE STAFF CONSOLE
smartclearance-console.web.app
onboarding clients, agents, triggers and the audit log

START AT THE FRONT DOOR



smartclearance.web.app: its bar leads to the Console and the Workspace; Book a demo and the film sit under the heading.

THE GUIDED DEMO · NO SIGN-IN



The route map: four lines into Munchly's workspace; every station is a button. Amber is a person's yes, red the bin.

TRY IT IN TWO MINUTES

1. **Open the guided demo** at smartclearance-demo.web.app: a route map of four lines, the console and three batches, each ending its own way.
2. **Pick a station, or Play the whole route** A laptop and a phone run the real screens; → moves on a beat, 1 to 4 jump to a line, M opens the map.
3. **End on the results** ₹21,152, ₹16,917 and ₹18,632 recovered, with 144 packs destroyed on a second yes.

FOR JUDGES

The workspace runs the journey live on Cloud Run and Gemini, with every person signing in: the next slide has who is who, the password and each step. Every company, person and figure is fictional; the money rules follow Journey Map v4.1.

Run the live demo: who is who, and each step

Smart-Clearance Silicon Prophets

Detailed demo setup, steps and sample data: `DEMO.md` **on GitHub**
`github.com/aibuilder2026/smart-clearance/blob/main/DEMO.md`
Password for all personas shared separately with `evaluator@hack2skill.com`

Console: **smartclearance-console.web.app**; everyone else: **munchly-smartclearance.web.app**. A private window for each person. Everyone below is fictional.



Neha Kulkarni · Smart-Clearance staff · console
`neha.kulkarni@smartclearance.example`
Resets the journey and its day length; fires runs and timers



Priya Deshmukh · Munchly Foods · supply chain
`priya.deshmukh@munchly.example`
Confirms Setup, approves the plan, reviews papers, the second yes



Rakesh bhai · Rakesh Traders · distributor, chips
`rakesh-traders@google.example`
Permission, label photo, Tally invoice, van round, any destruction



Lakshmi Agencies · distributor, Mango Drink
`lakshmi-agencies@google.example`
The Mango Drink's label photo; the staff sale on its UPI receipt



Ganesh ji · Shree Ganesh Kirana · one of 38
`shree-ganesh-kirana@google.example`
Orders from the Hindi scheme; each shop signs in as its own name



Agrawal ji · Agrawal Wholesale · ExpireSoon buyer
`agrawal-wholesale@google.example`
Bids on the lot from outside Munchly's territories



Meera · Feeding India · food bank
`feeding-india@google.example`
Collects the Mango Drink's donation and issues the receipt



Arjun Nair · Munchly Foods · workspace admin
`arjun.nair@munchly.example`
Users, guardrails, integrations, the audit log (optional)

THE LIVE JOURNEY, STEP BY STEP · MASALA CHIPS MF-2409-117

1. **Neha · console**
   Clients → Munchly Foods → Actions → "Reset journey…", pick Rehearsal (an hour a day) or Demo (five minutes). Day 0, 08:00.
2. **Priya · workspace**
   Setup: the stock export is already mapped, 8 of 8 columns. "Confirm and start watching".
3. **Rakesh · workspace**
   Today: "Allow", the one-time permission for the agents to act in his name.
4. **Neha · console**
   Agents → Watcher → "Run now": the chips and the Mango Drink are flagged at risk.
5. **Rakesh · workspace**
   Label photo: "Upload a photo" (the sample label below), then "Send photo". Gemini reads it against the record.
6. **Priya · workspace**
   Route Room: the Valuer's prices, the Router's plan. "Review and approve", then "Approve · release the agents".
7. **Each kirana · workspace**
   e.g. Ganesh ji: Offers → the scheme in Hindi → order. His tax invoice comes with the order.
8. **Agrawal ji · ExpireSoon**
   Marketplace → "View lot" → bid ₹13. The Negotiator counters at ₹14.20; "Accept", pay the token.
9. **Rakesh · workspace**
   "Load the buyer's truck", "Issue from Tally" (the IRN and signed QR), then "Start the round".
10. **Priya · workspace**
    Paperwork: the invoice, credit note, ITC memo and kiranas' invoices, signed, with PDFs. "Mark reviewed".
11. **Neha · console**
    Impact → "Expiry day · report" → "Report now": Impact posts the ledger and the batch clears.
12. **Priya and Rakesh**
    The Command Center reads ₹21,152 recovered; the Ledger shows money, GST and impact; Rakesh ends whole.

THE SLOW WEEK · STEPS 13 TO 18

At step 7, once two shops have ordered, Neha closes the scheme (Agents → Outreach → "Close now"): 540 packets stay at the godown. After "Report now", Rakesh's Today asks him to destroy them: two photos (the 540-pack samples), the agency, its certificate, the window and two witnesses; the officer, both witnesses and he sign in place. Priya: "Review the destruction", Vision's four checks, "Approve · issue the papers".

THE MANGO DRINK, ALONGSIDE

Lakshmi Agencies sends its label photo and records the staff sale on its UPI receipt; Meera collects the donation and issues the receipt.

DAY LENGTH

Real time 1,440 min · Rehearsal 60 · Demo 5 · Fast 1. Day 0 is always 2 Oct 2026, 08:00, so the chips have 47 days left. Set it back to Real time after a demo.

Sample files, in the public repository: the labels `design3/system/img/labels/MF-2409-117.webp` and `MF-2410-118.webp`; the staff receipt `design3/system/img/receipts/MF-2410-118.webp`; the evidence `frontend/workspace/tests/journey/evidence/MF-2409-117-before-540.webp` and `…-after-540.webp`.

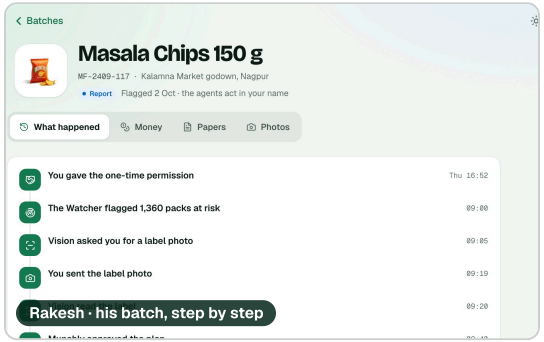
Three journeys, three endings

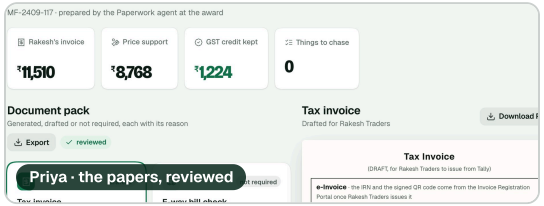
Smart-Clearance Silicon Prophets

The same ten agents on three batches, as the guided demo plays them and the live workspace runs them on Cloud Run and Gemini: every pack sold, three exits at once, and what happens when stock still doesn't sell.

JOURNEY 1 · MASALA CHIPS, NAGPUR

Every pack sold — ₹21,152 recovered

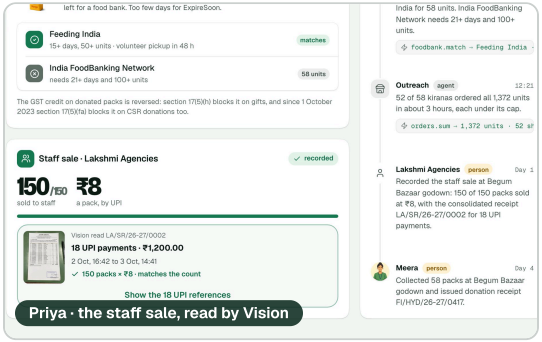


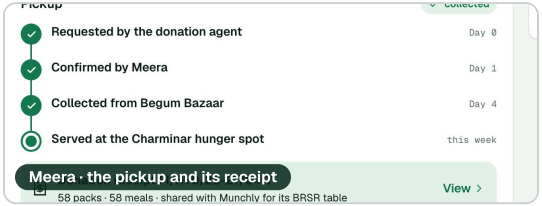


- 31 of 38 kiranas ordered 588 from the Hindi scheme
- Agrawal Wholesale took 772 at ₹14.20, after the Negotiator's counter
- INV/26-27/0931 with its IRN; CN/0117 for ₹8,768: Rakesh ends whole

JOURNEY 2 · MANGO DRINK, HYDERABAD

Three exits at once — ₹16,917 recovered

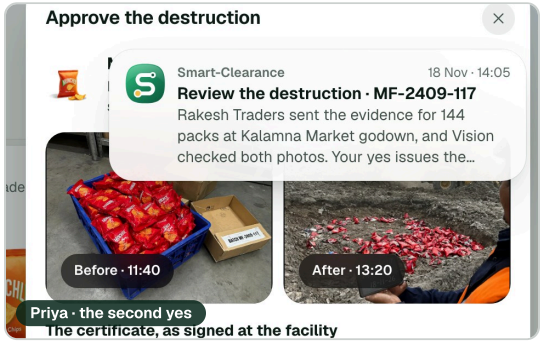




- 52 of 58 kiranas ordered 1,372; too few days left for ExpireSoon
- 150 to staff at ₹8: one receipt, 18 UPI payments, read by Vision
- 58 to Feeding India: receipt FI/HYD/26-27/0417, 58 meals

JOURNEY 3 · MASALA CHIPS, A SLOW WEEK

Destroyed, on proof — ₹18,632 recovered





- 23 shops ordered 444 of 588; one said Not this time
- 144 destroyed at the godown on two photos, Vision's checks and a certificate signed in place
- Priya's second yes issues CN/0118 for ₹3,542.40

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you



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